

Lecture 16: Maximum Likelihood Estimator (MLE) - Introduction

Let $X_1, \dots, X_n$ be a random sample from a population with pmf/pdf f_θ , $\theta \in \Theta$.

The joint pmf/pdf of $X_1, \dots, X_n$ is

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_{i=1}^n f_\theta(x_i)$$

The likelihood function $L(\theta | x_1, \dots, x_n) = \prod_{i=1}^n f_\theta(x_i)$, is a fn. of θ given the data $(x_1, \dots, x_n)$

(looking at the jt. pmf/pdf in a different way)

For a discrete distr., $L(\theta | x)$ is the probability for observing $x_1, \dots, x_n$, whereas, for a continuous distr., it is proportional to a probability statement (recall the jt. cdf, here).

The maximum likelihood estimation approach is to find θ , which maximizes the likelihood (or equivalently, probability) of a given set of $x_1, \dots, x_n$ (data).

Hence, $\hat{\theta}$ is an MLE of θ if $\hat{\theta} = \arg \max_{\theta \in \Theta} L(\theta | x)$.

NOTE 1. MLE is always a fn. of sufficient statistic

If T is sufficient, by NFFT, $f_\theta(x) = h(x)g_\theta(T(x)) = L(\theta | x)$

Maximizing $L(\theta | x)$ wrt $\theta \Leftrightarrow$ maximizing $g_\theta(T(x))$ wrt θ .

Hence etc.,

2. Since, $\log$ is a monotone fn., equivalent defn. would be

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} \log L(\theta | x)$$

(Most of the times it is more convenient to work with

$l(\theta | x) = \log L(\theta | x)$, the log likelihood fn.

3. Invariance property of MLE.

For $\theta \in \Theta$, if $\hat{\theta}$ is MLE, then $g(\hat{\theta})$ is MLE of $g(\theta)$, for any fn. $g(\cdot)$ (real valued fn.).

4. If $L(\theta|z)$ or $l(\theta|z)$ is differentiable wrt θ & the maximum is at an interior point ~~of~~ $\hat{\theta}$ of Θ and not a point on the boundary, then if $\frac{\partial l(\theta)}{\partial \theta} \Big|_{\hat{\theta}} = 0$ and $\frac{\partial^2 l(\theta)}{\partial \theta^2} \Big|_{\hat{\theta}} < 0$, $\hat{\theta}$ is an MLE of θ .

Multiparameter case $x_1, \dots, x_n$ r.s. from $f_{\underline{\theta}}$, $\underline{\theta} = (\theta_1, \dots, \theta_k) \in \Theta \subseteq \mathbb{R}^k$, we need to check, $\frac{\partial l(\underline{\theta})}{\partial \underline{\theta}} \Big|_{\hat{\underline{\theta}}} = \underline{0}_{k \times 1}$ and $\frac{\partial^2 l(\underline{\theta})}{\partial \underline{\theta} \partial \underline{\theta}'} \Big|_{\hat{\underline{\theta}}}$ is negative definite.
Incidentally, $\frac{\partial^2 l(\underline{\theta})}{\partial \underline{\theta} \partial \underline{\theta}'} = \left(\left(\frac{\partial^2 l(\underline{\theta})}{\partial \theta_i \partial \theta_j} \right) \right)$ is the $k \times k$ symmetric Hessian matrix.

Examples

1. $x_1, \dots, x_n$ r.s. from $\text{Bin}(1, \theta)$, $\theta \in [0, 1]$

$$L(\theta|z) = \theta^{\sum x_i} (1-\theta)^{n-\sum x_i}$$

$$l(\theta|z) = \sum x_i \ln \theta + (n - \sum x_i) \ln(1-\theta)$$

$$\frac{\partial}{\partial \theta} l(\theta|z) \Big|_{\hat{\theta}} = \frac{\sum x_i}{\hat{\theta}} - \frac{n - \sum x_i}{1 - \hat{\theta}} = 0 \Rightarrow \hat{\theta} = \frac{\sum x_i}{n} = \bar{x} \quad \left(\begin{array}{l} \text{is the max. likelihood} \\ \text{estimate, given the data} \\ \text{if the two der. conditions} \end{array} \right)$$

$$\frac{\partial^2 l(\theta)}{\partial \theta^2} \Big|_{\hat{\theta}} = -\frac{\sum x_i}{\hat{\theta}^2} - \frac{(n - \sum x_i)}{(1 - \hat{\theta})^2} < 0$$

Hence, $\hat{\theta} = \frac{\sum_{i=1}^n x_i}{n}$ is MLE of θ .

(What if $\theta \in (0, 1)$?
Essentially, saying Θ is different from the above case)

2. $x_1, \dots, x_n$ r.s. from $P(\theta)$, $\theta \geq 0$.

$$L(\theta) = e^{-n\theta} \frac{\theta^{\sum x_i}}{\prod_{i=1}^n x_i!}, \quad l(\theta) = -n\theta + \sum x_i \ln \theta - \ln(\prod x_i!)$$

$$\frac{\partial}{\partial \theta} l(\theta) \Big|_{\theta=\hat{\theta}} = -n + \frac{\sum x_i}{\hat{\theta}} = 0 \Rightarrow \hat{\theta} = \frac{\sum x_i}{n} = \bar{x}$$

$$\frac{\partial^2}{\partial \theta^2} l(\theta) \Big|_{\theta=\hat{\theta}} = -\frac{\sum x_i}{\hat{\theta}^2} < 0$$

$\Rightarrow \hat{\theta} = \frac{\sum x_i}{n}$ is MLE of θ .